AI/ML Internship - Task 1

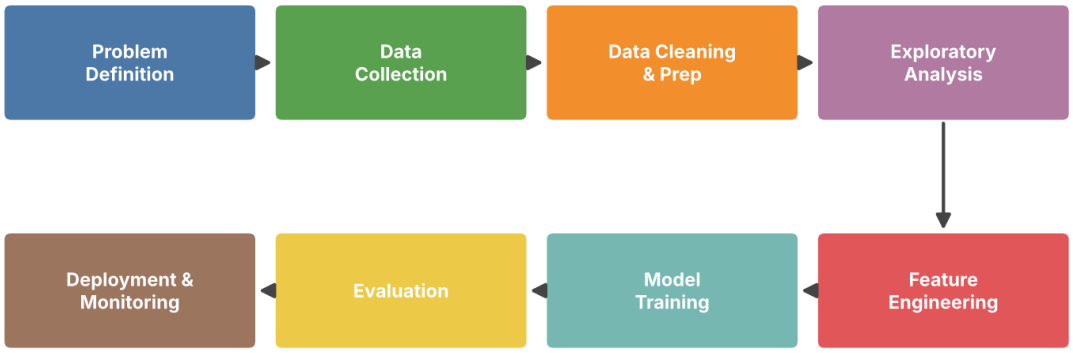
Development Environment Setup, Python Foundations & Exploratory Data Analysis

Report summarising the EDA findings for the Student Performance Data Explorer, the Iris dataset challenge, and the initial Employee Attrition analysis.

1. Environment and workflow

- Python 3, VS Code, Git and Jupyter Notebook with an isolated virtual environment (`python -m venv venv`).
- Libraries: NumPy, Pandas, Matplotlib, Seaborn (see `requirements.txt`).
- Project organised with data / notebooks / src / reports / images / models folders and version-controlled with Git.

AI/ML Project Lifecycle



Iterate: monitor drift, retrain and improve

diagram ml lifecycle

Project Folder Structure

```
AI-ML-Internship/
  data/                 raw and cleaned CSV datasets
  notebooks/            Jupyter notebooks (.ipynb)
  src/                  reusable Python scripts
  reports/              PDF report and EDA write-ups
  images/               charts, diagrams, screenshots
  models/               saved models (used from Task 2)
  requirements.txt      pinned dependencies
  README.md             project documentation
  .gitignore            files excluded from Git (venv/ etc.)
```

diagram folder structure

Exploratory Data Analysis Workflow

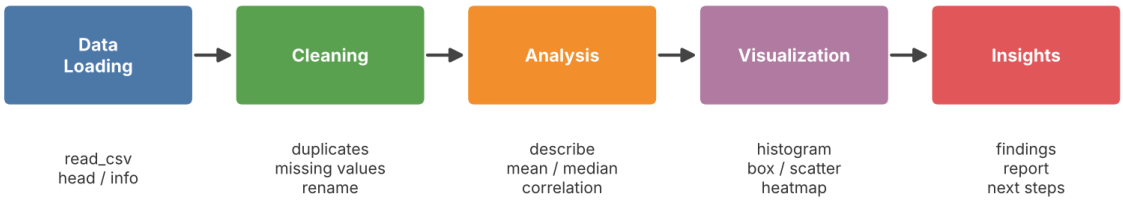
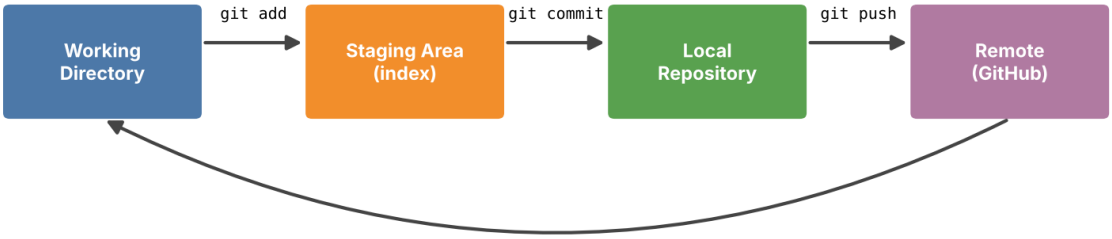


diagram eda workflow

Git & GitHub Workflow

git init / git clone → start here



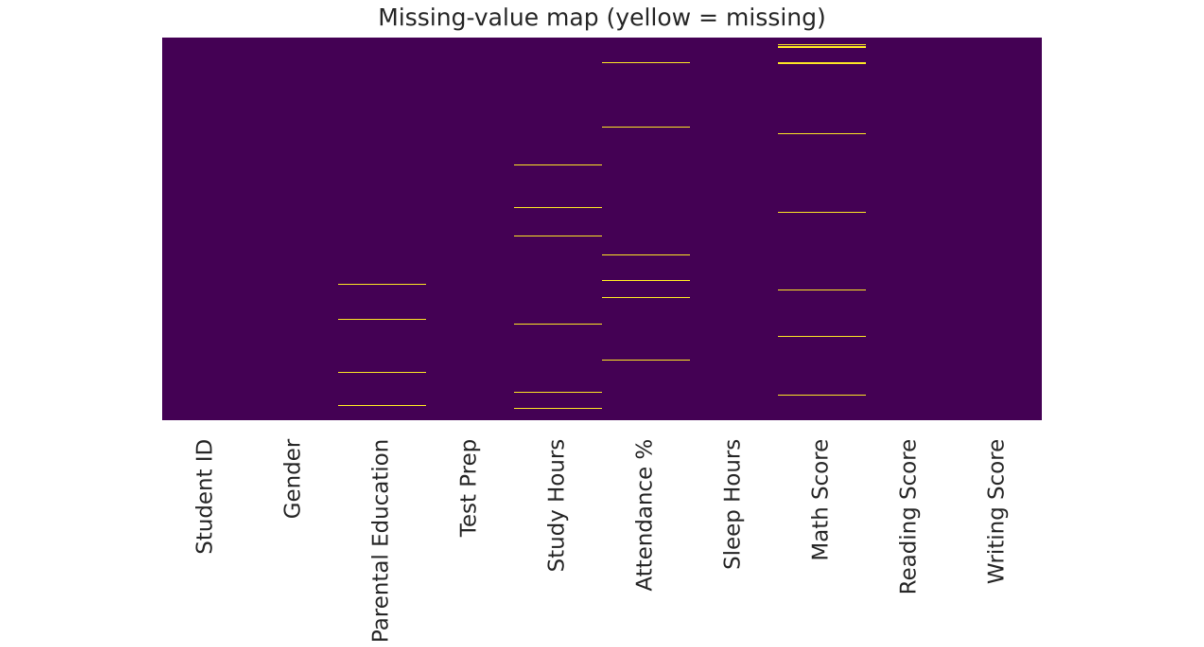

```
git pull / git fetch  (sync remote changes back to local)
```

diagram git workflow

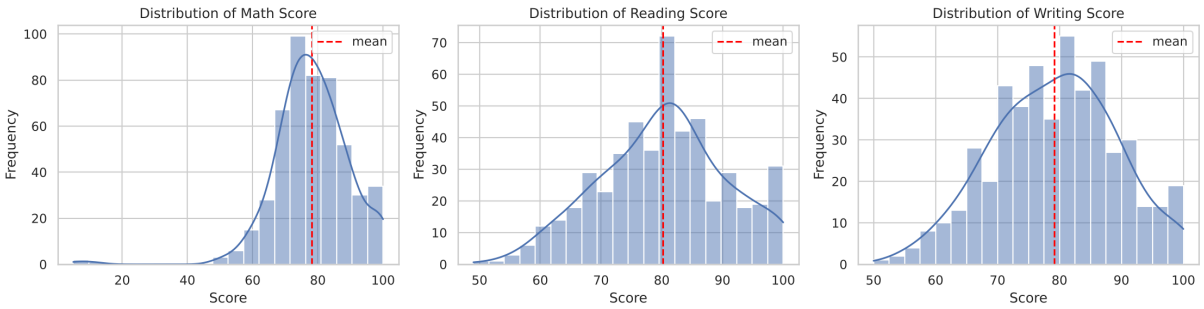
2. Student Performance Data Explorer

Dataset: 509 rows x 10 columns (synthetic, reproducible via `src/generate_datasets.py`). **Data quality:** 9 duplicate rows and 36 missing cells were found. Duplicates were removed; numeric gaps filled with the median and categorical gaps with the mode. Cleaned shape: 500 x 10.

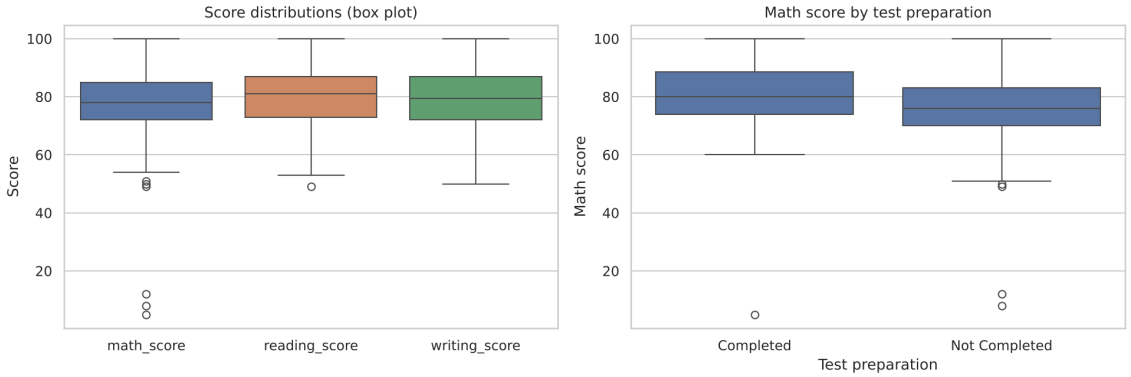
Score	Mean	Median	Std Dev
math_score	78.14	78.0	11.54
reading_score	80.2	81.0	10.46
writing_score	79.16	79.5	10.06



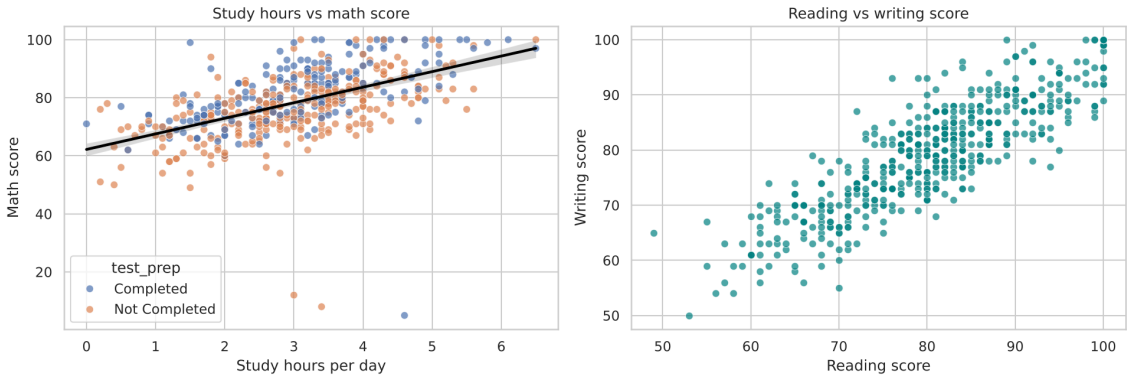
student missing values



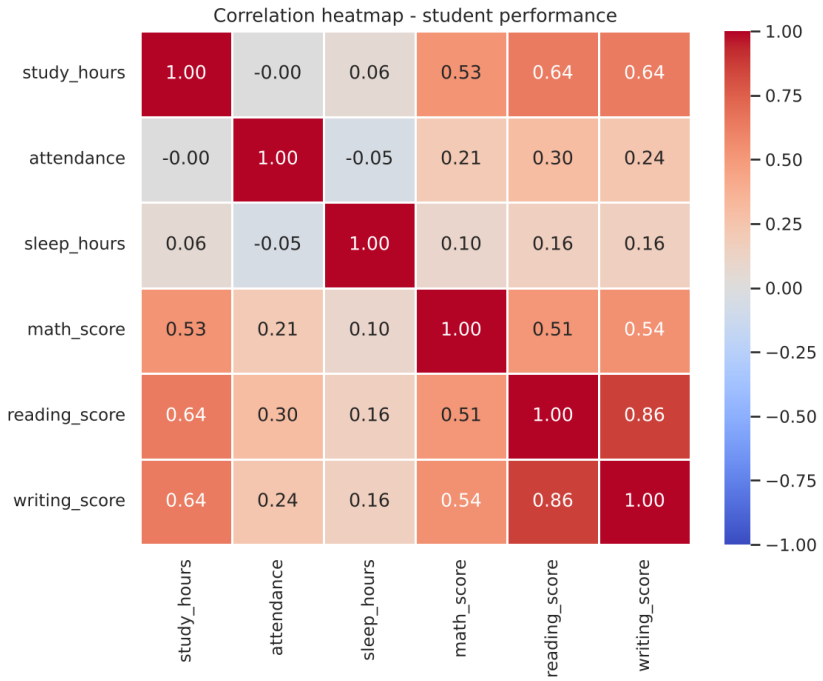
student histograms



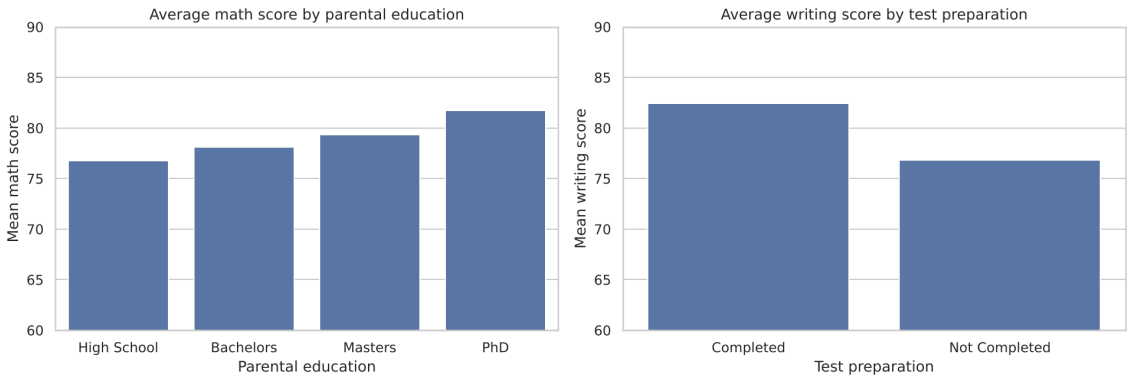
student boxplots



student scatter



student correlation heatmap



student group bars

Key findings

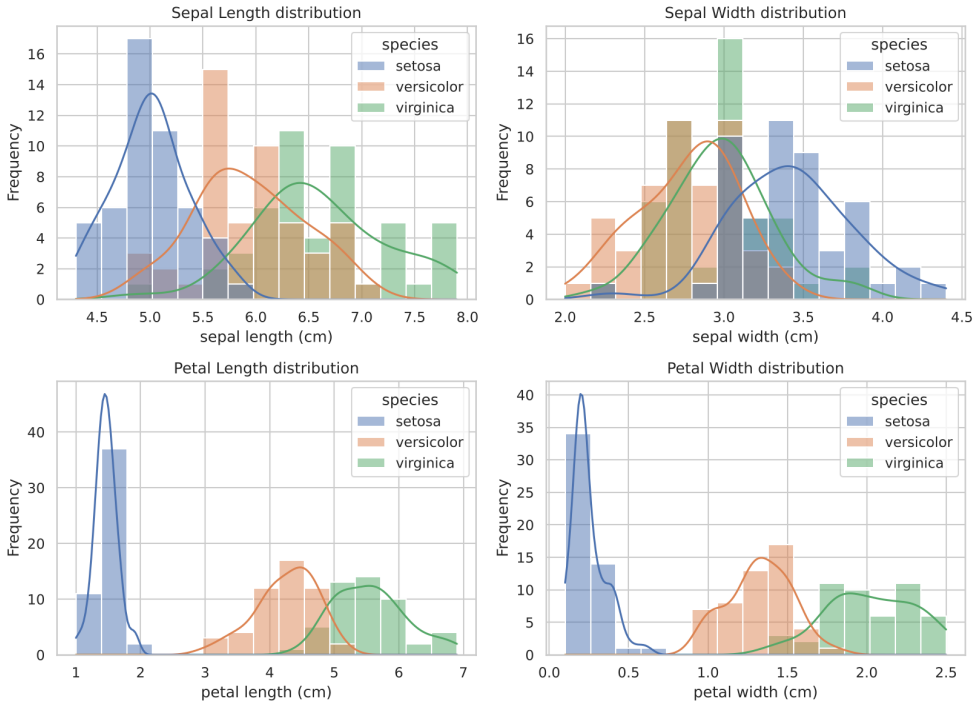
- Study hours is the strongest driver of marks (correlation with math 0.53, reading 0.64).
- Reading and writing scores are tightly linked (r = 0.86).

- Attendance has a smaller positive effect (r = 0.21 with math); sleep hours are weak (r = 0.10).
- Students who completed test prep average 81.3 in math vs 75.9 without it.
- Box plots show a handful of extreme math scores (both very low and perfect), flagged by the 1.5 x IQR rule.
- The gender gap in average scores is negligible.

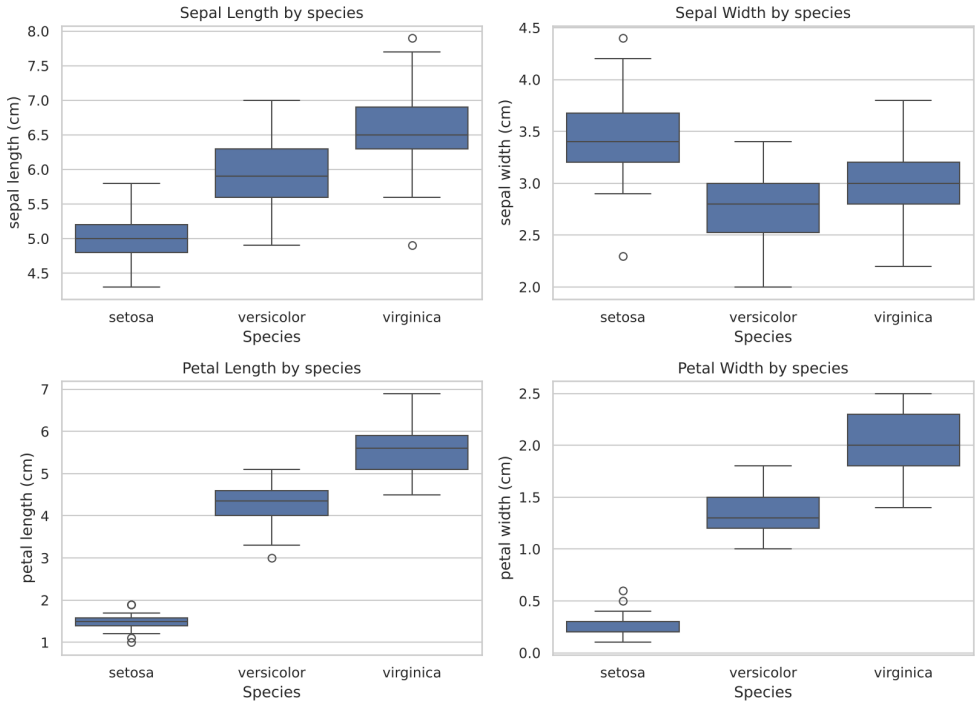
3. Iris dataset - advanced challenge

150 observations, 4 numeric features and 3 balanced species. No missing values; 1 duplicate row removed. Mean measurements per species:

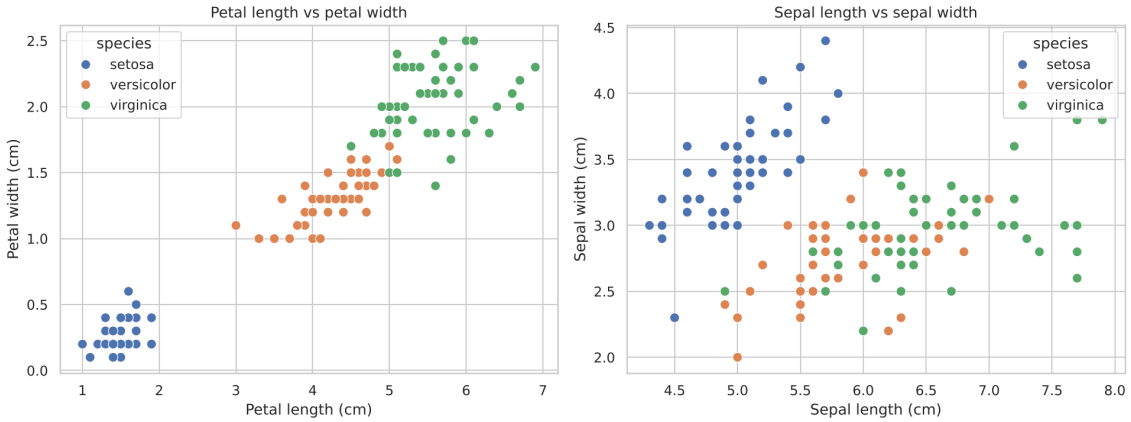
species	sepal length	sepal width	petal length	petal width
setosa	5.01	3.43	1.46	0.25
versicolor	5.94	2.77	4.26	1.33
virginica	6.59	2.97	5.55	2.03



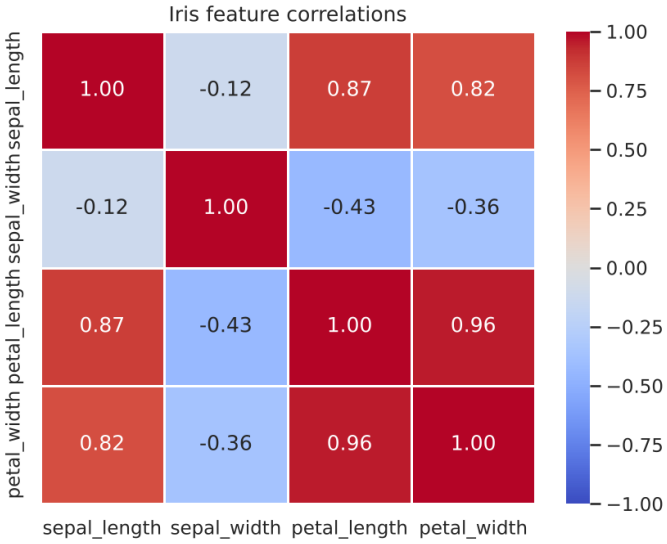
iris histograms



iris boxplots



iris scatter



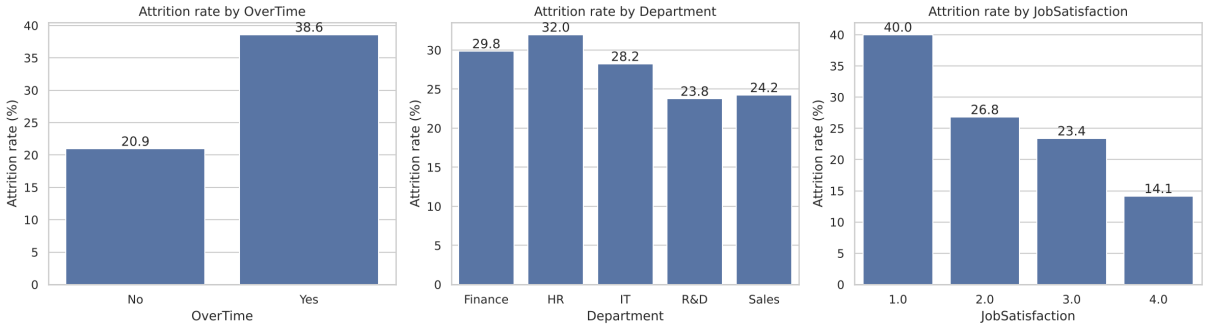
iris correlation heatmap

Key findings

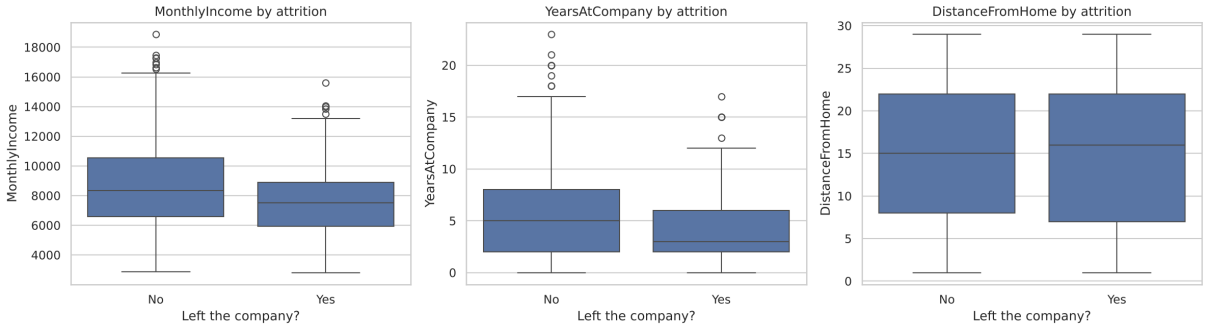
- Petal length and petal width are almost perfectly correlated (r = 0.96).
- Setosa is clearly separable on petal size; versicolor and virginica overlap in sepal measurements but separate well on petals.
- Sepal width is the weakest feature (r = -0.43 with petal length).
- Petal measurements are the best candidate features for a classifier in Task 2.

4. Capstone progress - Employee Attrition (initial EDA)

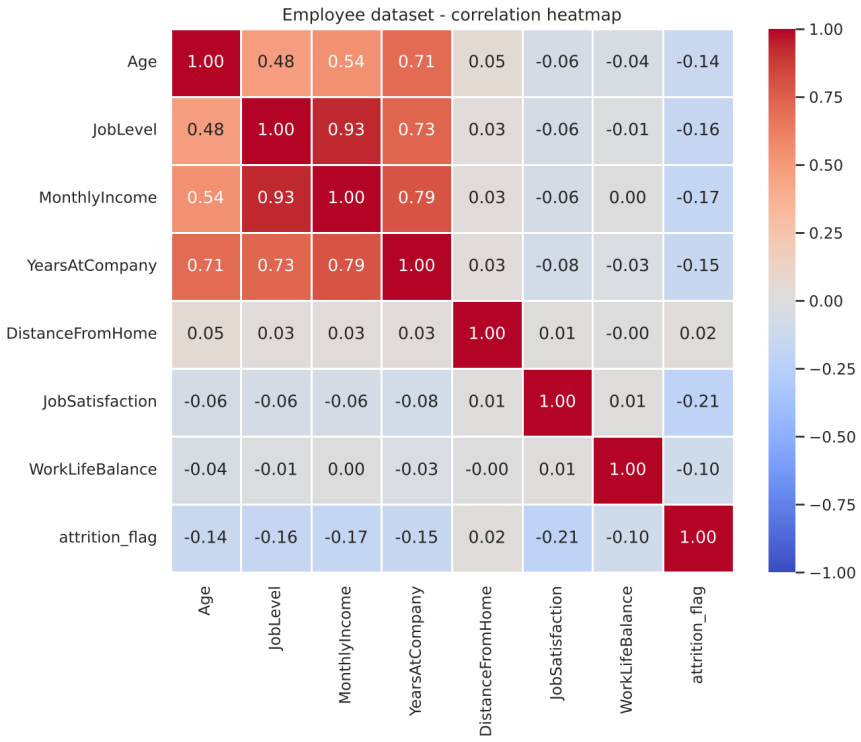
1200 employees; overall attrition rate 26.1% (imbalanced target). Missing income values were filled with the job-level median and missing satisfaction with the median. *The dataset is synthetic - swap in the real HR file when available.*



employee attrition by category



employee numeric boxplots



employee correlation heatmap

Initial observations

- Overtime is the clearest signal: 38.6% attrition with overtime vs 20.9% without.
- Lower job satisfaction means higher attrition (40.0% at level 1 vs 14.1% at level 4).
- Income (r = -0.17) and tenure (r = -0.15) are negatively correlated with attrition.
- Commute distance shows almost no linear relationship (r = 0.02).
- Age, tenure, job level and income are inter-correlated, so watch multicollinearity when modelling.

5. Conclusion and next steps

The environment, Python fundamentals, data cleaning and visualisation workflow are in place and documented. Task 2 will use the cleaned datasets to preprocess features, split data, train baseline models (logistic regression, random forest) and evaluate them using precision, recall and F1.